

**Project Design Phase-II**  
**Solution Requirements (Functional & Non-functional)**

|               |                                                                                          |
|---------------|------------------------------------------------------------------------------------------|
| Date          | 03 July 2026                                                                             |
| Team ID       |                                                                                          |
| Project Name  | Comprehensive Analysis Dietary Strategies with Tableau:A college food choices case study |
| Maximum Marks | 4 Marks                                                                                  |

**Functional Requirements:**

Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic)         | Sub Requirement (Story / Sub-Task)                                                                                                              |
|--------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1   | Data Collection & Preparation         | Collect the College Food Choices dataset, connect the dataset to Tableau, validate records, and prepare data for analysis.                      |
| FR-2   | Data Preparation                      | Handle missing values, clean and format data, verify data types, and prepare fields for visualization.                                          |
| FR-3   | Data Visualization                    | Create charts to analyze gender, breakfast habits, food choices, comfort food reasons, exercise, vitamins, and healthy feeling.                 |
| FR-4   | Dashboard                             | Create an interactive dashboard, display key insights, use Gender, Breakfast, and Healthy Feeling filters, and support interactive exploration. |
| FR-5   | Story                                 | Create a 5-scene Tableau Story, present major insights in sequence, and highlight dietary and lifestyle patterns.                               |
| FR-6   | Performance Testing                   | Verify data rendered, utilization of filters, calculated fields, and the number of visualizations/graphs.                                       |
| FR-7   | Web Integration                       | Publish the Tableau dashboard and embed the dashboard and Story using Flask with a responsive interface.                                        |
| FR-8   | Project Demonstration & Documentation | Record the project explanation video and prepare step-by-step project documentation.                                                            |

**Non-functional Requirements:**

Following are the non-functional requirements of the proposed solution.

| FR No. | Non-Functional Requirement | Description                                                                              |
|--------|----------------------------|------------------------------------------------------------------------------------------|
| NFR-1  | Usability                  | The Tableau dashboard and visualizations should be easy to understand and interact with. |
| NFR-2  | Security                   | The analysis should accurately represent the College Food Choices dataset.               |

|       |                     |                                                                                                        |
|-------|---------------------|--------------------------------------------------------------------------------------------------------|
| NFR-3 | <b>Reliability</b>  | The dashboard, filters, and visualizations should function consistently and provide reliable insights. |
| NFR-4 | <b>Performance</b>  | The Tableau dashboard and worksheets should load and respond efficiently.                              |
| NFR-5 | <b>Availability</b> | The published dashboard should be accessible to intended users when deployed.                          |
| NFR-6 | <b>Scalability</b>  | The solution should support larger datasets and additional dietary or lifestyle variables in future.   |